



National distances, international experience, and venture capital investment performance[☆]



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ARTICLE INFO

Article history:

Received 22 February 2011

Received in revised form 19 July 2013

Accepted 17 August 2013

Available online 9 November 2013

Field Editor: D. Jennings

Keywords:

Venture capital

Institutional distance

Cultural distance

International experience

Exit

ABSTRACT

This study examines how the performance of cross-border venture capital investments is affected by national institutional and cultural distances between the environments of venture capitalists (VCs) and investee ventures. We propose that institutional and cultural distances will decrease VCs' effectiveness in conducting venture capital activities and negatively affect investment performance in terms of exit success, and obtain supportive evidence while controlling for geographic distance. We further analyze how VCs can use their international experience to mitigate the negative consequences of national distances. We find that while broad international experience in diverse countries attenuates the deleterious effects of institutional distance in a significant way, it does not have a similar impact in attenuating the negative effects of cultural distance.

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1. Executive summary

This study examines how the performance of international venture capital investments is affected by the distances in national institutional and cultural environments between venture capitalists (VCs) and investee ventures. The increased internationalization of the venture capital sector has triggered the interest of entrepreneurship scholars in studying how distinct national institutions and cultures impact key entrepreneurial activities and venture capital financing. We extend existing literature by addressing the impact of two distinct but complementary national distances on exit success of cross-border venture capital investments: institutional and cultural distances. While the cultural distance perspective captures the impact of national differences in cognitive patterns and values, the institutional distance perspective captures the impact of differences in regulatory systems as well as in institutionalized practices, beliefs and assumptions that govern and order business relationships. Cultural distance raises barriers to information sharing, reduces trust, and increases transaction costs. Institutional distance increases the risk of legitimacy loss, sanctions, and additional transaction costs associated with incongruence of institutionalized practices. We thus hypothesize that both cultural and institutional distances will have a negative impact on the likelihood of a successful exit in the form of venture IPO or sale.

[☆] The authors contributed equally to this project. We are grateful for the constructive guidance and insightful comments provided by the field editor, Dev Jennings. We also thank the anonymous reviewers, Brian Silverman and Akie Iriyama for their invaluable suggestions. We acknowledge the financial support from the Social Sciences and Humanities Research Council of Canada. The usual disclaimer applies.

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The high costs of institutional and cultural distances have raised the issue of whether VCs can mitigate national distances by gaining cross-border investment experience. We address two types of investment experience. While broad international experience reflects the breadth and scope of experience in terms of the diversity of countries in which a VC has invested, host country-specific experience represents the depth or intensity of investment in the host country. We posit that both types of cross-border experience can reduce the adverse effects of institutional and cultural distances on exit success. While broad international experience in diverse countries increases awareness of national differences and allows VCs to develop adaptive capabilities to translate experiences gained from one environment to another, host country experience allows for deeper acquisition of institutional and cultural knowledge and the opportunity to perfect VCs' routines in a particular host country.

We further expect that the ease with which VCs can develop their international capabilities to mitigate negative consequences of institutional and cultural distances varies. Institutions tend to be more explicit and reflected more strongly in practices that are salient and susceptible to change. In contrast, values, beliefs, and cognitive patterns that are carried through culture tend to be more tacit and more impervious to change. Further, compared with behaviors incompatible with local cultures, behaviors incompatible with local institutions are more likely to be formally punished or endure social sanctions, which propel VCs to learn from experience. We thus propose that international experiences have more salient mitigating effects on the adverse impact of institutional distance than on the impact of cultural distance. To our best knowledge, the current study represents one of the first attempts to examine the different roles of experience in mitigating costs arising from both institutional and cultural distances.

We test our arguments by using a large sample of cross-border venture capital investments between 1996 and 2008 and controlling for geographic distance. Both logit and Heckman analyses show that institutional and cultural distances negatively affect the likelihood of exit success. Further, broad international experience alleviates the negative effect of institutional distance but neither VCs' broad international experience nor VCs' host country-specific experience mitigates significantly the negative impact of cultural distance.

Concerning the insignificant mitigating effects of international experience with regard to *cultural* distance, we suggest that the episodic nature of VCs' engagement in the host countries may not provide sufficient exposure to support deep learning that is required to bridge the cultural gap. Moreover, in many cases the perception of dissimilarity and thus the liability of outsidership cannot be completely eliminated because of attributes that are difficult to change (e.g., differences in physical features).

Our study has several implications for practice. First, distances matter to performance, but the main challenges VCs face may not arise from physical or geographic distance, as often discussed in the current literature, but from institutional and cultural distances. Second, our analysis suggests that VCs need to be aware of the differences in acquiring institutional and cultural knowledge. When recruiting professional staff members, VCs might want to prioritize the possession of diverse international experience by candidates, and ensure that candidates can adapt to changes in local environments so that the knowledge they possess remains current.

2. Introduction

Venture capitalists (VCs) play a critical role in providing not only scarce financial resources but also value-added services to high-potential entrepreneurial ventures (Baum and Silverman, 2004; Busenitz et al., 2004; Dimov and Shepherd, 2005; Hellmann and Puri, 2002; Sapienza, 1992). From its origins in the U.S., the venture capital industry has expanded around the world. Recent years have witnessed an increase in the number of venture capital deals across national borders (Aizenman and Kendall, 2008; Kenney et al., 2008; Li, 2012; Wright et al., 2005). Through international investments, VCs are able to tap into growth opportunities in foreign markets (Dai et al., 2012; Guler and Guillen, 2010). They may, however, face significant liability of foreignness in international investment because of the challenges arising from geographic, cultural, and institutional distances between home and host countries (Zaheer, 1995).

The current study examines how the performance of international venture capital investments is affected by the distances in national institutional and cultural environments between VCs and investee ventures. The idea that national distances affect the conduct and performance of businesses operating across borders has been at the core of international business research for decades (see, for example, the seminal works of Hymer, 1960; Johanson and Vahlne, 1977). Recently, the increased internationalization of the venture capital sector has triggered the interest of entrepreneurship scholars in studying how distinct national institutions and cultures impact key entrepreneurial activities and venture capital financing (e.g., Bruton et al., 2009; Bruton et al., 2005). For instance, Dai et al. (2012) have investigated the effect of cultural distance on VCs' cross-border investment behavior and performance in Asia. The current study extends existing literature in three important ways.

First, we analyze the effect of institutional distance on VCs' cross-border performance in addition to the effect of cultural distance, the focus of prior studies (Dai et al., 2012). While the cultural distance perspective captures the impact of national differences in cognitive patterns and values (Hofstede, 1980; Kogut and Singh, 1988; Shenkar, 2001), the institutional distance perspective captures the impact of differences in regulatory systems as well as in institutionalized practices, beliefs and assumptions that govern and order business relationships (Peng and Heath, 1996). Our study thus provides economic and sociological insights to complement the social-psychological insights derived from the cultural distance perspective. Furthermore, we argue that institutional distance influences the effectiveness of VCs' primary activities and the performance of cross-border investments through mechanisms that are distinct from those that are triggered by cultural distance. Specifically, while cultural distance leads to rising barriers to information sharing, reduced trust, and increased transaction costs, institutional distance increases the risk of legitimacy loss, sanctions, and additional transaction costs associated with incongruence of institutionalized practices. Incorporating the two distinct but complementary perspectives of national distance in the theoretical

framework thus allows us to provide a fuller account for the impact of national distance on venture capital investment performance.

Our second contribution is to investigate how VCs' international experience can help mitigate the negative consequences of institutional and cultural distances. Scholars have only recently paid attention to the role of international experience in VCs' cross-border investment strategies such as location choice and syndicate partner selection (Guler and Guillen, 2010; Meuleman and Wright, 2011). Informed by insights from organizational learning theories, we propose that cross-border experiences, including both broad international and host country specific experiences, can reduce the adverse effects of institutional and cultural distances on venture capital investment performance. We further posit that the ease with which VCs can develop their international capabilities to mitigate negative consequences of institutional and cultural distances varies. While institutions tend to be more explicit and reflected more strongly in practices that are salient and susceptible to change, values, beliefs, and cognitive patterns that are carried through culture tend to be more tacit and more impervious to change (Hofstede, 1989; Kostova, 1999; Kostova and Zaheer, 1999; North, 1990; Xu and Shenkar, 2002). Moreover, compared with behaviors incompatible with local cultures, behaviors incompatible with local institutions are more likely to receive salient legal or social sanctions, which propel VCs to learn from experience. We thus propose that international experiences have more salient mitigating effects on the adverse impact of institutional distance than on the impact of cultural distance. To our best knowledge, the current study represents one of the first attempts to examine the different roles of experience in mitigating costs arising from both institutional and cultural distances.

Our third contribution is to extend the study of the impacts of national distances on VCs' international investment performance to the global level. This extension is necessary since many VCs invest sequentially and sometimes simultaneously in different regions of the world. Analyses based on regional data (e.g., Dai et al., 2012) would be inadequate in such cases to account fully for the effects of international experience.

We test our theoretical arguments using a large sample of cross-border venture capital investments between 1996 and 2008 and obtain supportive evidence. Controlling for venture nation and industry fixed effects as well as geographic distance, we find that both institutional and cultural distances have negative effects on the likelihood of a successful exit in the form of venture IPO or sale. Further, broad international experience alleviates the negative effect of institutional distance on investment performance but does not mitigate significantly the negative impact of cultural distance.

3. Theory and hypotheses

3.1. Theoretical framework

As an increasing share of business activities are conducted across borders, national distance has become a key concept in management (Berry et al., 2010). Despite an almost universal recognition of the multi-dimensional nature of distance, scholars have used Kogut and Singh's (1988) cultural distance index as "the proxy of choice for national difference" (Berry et al., 2010; Xu and Shenkar, 2002: 608). At the core of the cultural distance perspective is an emphasis on (1) the cognitive nature of culture, defined as "the collective programming of the mind that distinguishes members of one category of people from another category" (Hofstede and Bond, 1988: 614); and (2) work related values such as acceptance of authorities and tolerance of ambiguity and risk (Hofstede, 1980). This perspective captures a critical aspect of national distance and accounts for some important differences between managers of the VCs and those of the investee ventures.

Nevertheless, the cultural distance perspective "does not capture the complexity of cross-country differences" (Xu and Shenkar, 2002: 608). Important national distance dimensions that cultural distance fails to fully capture include differences in the regulatory systems, and differences in the institutionalized practices, beliefs and assumptions about production, exchange, and distribution. To incorporate aspects of national distance related to the institutional environment, scholars have proposed an institutional distance perspective (e.g., Peng and Heath, 1996) that has its roots in both economics and sociology (North, 1990; Scott, 2001). While "sociologists focus on the legitimacy-defining role that institutions take on" and emphasize the role of institutions in shaping how people in a society think and behave (Peng and Heath, 1996: 499), economists emphasize the role of the institutional environment in regulating and shaping economic activities "by providing the set of fundamental political, social and legal ground rules that establishes the basis for production, exchange and distribution" (Davis and North, 1971: 6).

One of the key conceptualizations of institutional differences across countries is developed by Henisz and Williamson (1999) and La Porta et al. (1998). It focuses on national governance systems (Berry et al., 2010), which include the regulatory systems, the accountability, safeguards, incentives and dispute resolution processes that the regulatory systems legitimize, and the rules that govern exchange relations and order the activities of business stakeholders. We apply this conceptualization to capture the key differences in the national institutional environments with regard to the action arenas and task domains of VCs, investee ventures, and their interactions.

Consequently, we incorporate in our theoretical framework two complementary conceptualizations of national distance: the cultural distance perspective with a focus on the cognitive and value dimensions of culture, and the institutional distance perspective with a focus on national governance institutions. Below, we discuss the activities of VCs and explore how cultural and institutional distances affect the effectiveness of these activities and thus the performance of international investments. We then explore how VCs' cross-border investment experience might mitigate the effects of national distances.

3.2. The action arenas and task domains of VCs

Venture capital is a critical source of financial resources for entrepreneurial ventures that are usually credit constrained (Evans and Leighton, 1989). Unlike passive portfolio investors such as banks (de Bettignies and Brander, 2007; Winton and Yerramilli, 2008), VCs seek to add value to entrepreneurial ventures beyond the money supplied (Arthurs and Busenitz, 2006; Barney et al., 1996; Bruton and Ahlstrom, 2003; Busenitz et al., 2004; Dimov and Shepherd, 2005; Edelman, 2002; Fried et al., 1998; Sapienza, 1992; Zarutskie, 2010). The primary non-financial activities that VCs perform to create value include (1) screening and deal selection, that is, identifying and assessing investment opportunities and selecting projects for investing (Brander et al., 2002; Gorman and Sahlman, 1989; Shepherd, 1999; Sorenson and Stuart, 2001); (2) contracting, that is, negotiating and renegotiating contracts to protect investors' rights (Baum and Silverman, 2004; Gompers, 2002; Li and Zahra, 2012); (3) monitoring investee performance (Gompers, 1995); and (4) advising/coaching, that is, providing value-added services to ventures such as strategy formulation, administrative support, certification, and attracting professional managers (Bygrave and Timmons, 1992; Fitza et al., 2009; Hellmann and Puri, 2000, 2002; Hsu, 2004; Kaplan and Stromberg, 2001; Megginson and Weiss, 1991; Stuart et al., 1999). We posit that the effectiveness of these activities, among other things, determines the performance of international venture capital investments. We now examine how national distances affect the effectiveness of these activities and thus investment performance.

3.3. Cultural distance and venture capital investment performance

Cultural distance is most often defined as the Euclidean distance between two countries in the space of four Hofstede cultural dimensions: Individualism–Collectivism, Power Distance, Uncertainty Avoidance, and Masculinity–Femininity (Kirkman et al., 2006; Kogut and Singh, 1988).³ Critics of this cultural distance perspective suggest that within country differences might exceed differences between countries. Nevertheless, prior studies have shown that differences in national cultures are significant and affect the ways people act and interact with each other (Reus and Lamont, 2009).

In the venture capital context, cultural distance negatively affects VC performance in three related ways. First, cultural distance creates communication problems, which hinder information sharing, a fundamental element to a successful VC–investee relationship (Amit et al., 1998; Dai et al., 2012). The effectiveness of VC–investee communications might be hindered by, for example, fundamental differences in communication styles (e.g., implicit communications in collectivistic societies versus explicit communications in individualistic ones), communication patterns (e.g., communications flow along hierarchical channels in high power distance cultures but are more diffused in low power distance cultures), and attention patterns (e.g., more versus less attention to the more powerful) (Thomas, 2001).

Second, cultural distance indicates dissimilarity in values and beliefs between VCs and investee ventures. Such dissimilarity can lead to different approaches towards deal selection, contracting, monitoring and advising, increase VCs' transaction costs and lower their effectiveness in conducting primary activities. For instance, as VCs from collectivist societies tend to rely more on “in-group” personal relationships to identify deals, they will encounter significant challenges in individualist societies where firms rely more on arm's-length market transactions to identify deals (Hofstede, 2001; Steensma et al., 2000). Similarly, VCs from the U.S. or the UK that place less emphasis on personal networks of entrepreneurs might choose suboptimal deals in Continental Europe or Asia where entrepreneurs' networks are more critical in the success of a venture (Bruton et al., 2005). In providing advice, VCs in the U.S. may have confrontations with the CEOs of the investee ventures on a regular basis (Fried and Hisrich, 1995); such a direct and confrontational approach might, however, result in destruction of relationships in China where advice to CEOs is often provided in a “face saving” manner through, for example, employment of intermediaries who know both the VCs and CEOs (Ahlstrom and Bruton, 2006; Bruton and Ahlstrom, 2003). In addition, while some societies with high uncertainty avoidance deal with uncertainties by establishing a highly formalized and hierarchical system (Hofstede, 2001; Steensma et al., 2000), others with low uncertainty avoidance prefer flexible structures that leave much room for improvisation and negotiation (Hofstede, 2001; Phillips and Wright, 1977). In this case, investees with low uncertainty avoidance orientation are not likely to internalize the logic of rigid rules and hierarchy as means of uncertainty reduction.

Third, cultural distance can lead to the “liability of outsidership” for VCs (Johanson and Vahlne, 2009; Vaara et al., 2012), which increases social discrimination and decreases trust level, thus lowering VC performance. Sociologists have noted that homophily is one of the most powerful mechanisms that influence the formation of social relationships, that is, people like to form relationships with people of similar values and beliefs (Sorenson and Stuart, 2001). Similarity tends to lead to attraction and trust, while dissimilarity tends to constrain the formation of relationships and trust (Terry, 2001; Vaara et al., 2012; Van Knippenberg

³ The first dimension, Individualism–Collectivism, relates to the nature of the social framework and the ties that members form within the framework. In individualistic cultures the ties between individuals and groups are loose, and people are supposed to take care of themselves and their immediate families only, while in collectivistic societies, the social framework is tighter, and members form strong ties within groups and distinguish in-groups and out-groups expecting their in-groups to take care of them and in exchange for that they owe absolute loyalty to the group (Hofstede, 2001). The second dimension, Power Distance, reflects the extent to which people accept the authority of those more powerful and consider the unequal distribution of power in organizations and the society legitimate. The third dimension, Uncertainty Avoidance, reflects the extent to which people in society feel threatened by situations that involve ambiguities and uncertainty and thus prefer to have clear and formal rules, receive unambiguous information and communications and place high value on data-rich expert analysis. The fourth dimension is Masculinity–Femininity. High masculinity implies valuing assertiveness and a focus on achievement, while in feminine cultures, quality of life and nurturing are dominant values.

and Van Leeuwen, 2001). Since trust is the key to successful collaboration, cultural distance will likely increase incidence of conflicts and decrease VCs' ability to minimize their exposure to investee opportunism (Gulati, 1995). In addition, given that trust is the foundation for a healthy climate of information exchange between VCs and investees, cultural distance hampers information gathering that is necessary for deal selection, monitoring and advising (Park et al., 2012).

In summary, cultural distance between VCs and investee ventures implies different values, beliefs and practices. These differences increase information and transaction costs, create liabilities of outsidership, and reduce the effectiveness of deal selection, contracting, monitoring, and advising. Given the important role of these activities in the success of venture capital investments, we expect that, other things being equal, a venture capital investment that is characterized by a high level of cultural distance will less likely have a successful exit.

Hypothesis 1. The larger the cultural distance between VCs and investee ventures, the lower the likelihood of exit success in international venture capital investments.

3.4. Institutional distance and venture capital investment performance

National governance institutions represent human constraints on the behavior of individuals and organizations (North, 1990). Countries differ widely in their institutional environments with regard to protecting property rights, enforcing contracts, and facilitating economic transactions. Such variation in national institutions is typically reflected in the nature and scope of the political, social and legal rules, the quality of enforcement, and the means of enforcement (e.g., formal judicial and enforcement processes versus social networks and social sanctions).

Institutional distance increases the incongruence between VCs' practices and local institutionalized practices in respect to deal selection, contracting, monitoring, and advising. For example, in countries with strong institutional environments (e.g., strong laws and accounting rules), VCs can rely on financial and accounting information to evaluate proposals and assess investment risk (Cumming et al., 2010; Manigart et al., 1997). In countries with weak institutional environments, however, the availability and accuracy of information are problematic, and VCs need to build personal contacts with relevant individuals who can provide information on entrepreneurs and their businesses (Bruton and Ahlstrom, 2003).

Regarding contract enforcement and monitoring, in countries with strong legal protection of investors, VCs commonly depend on complex, state-contingent contracts (e.g., shifting control rights in accordance with the performance of the investees) to protect their rights and minimize agency costs (Guler and Guillen, 2010; Lerner and Schoar, 2005). VCs can also rely on board memberships to monitor ventures, request formal financial reports, and place contractual restrictions on managers (Fried et al., 1998; Sapienza, 1992). These approaches, however, are less effective in countries with weak legal enforceability and weak protection of investors. In these countries, building networks and relationships is often considered a socially desirable approach to gather information, enforce contracts, and monitor ventures (Bruton and Ahlstrom, 2003; Bruton et al., 2005; Cumming et al., 2010). Similarly, VCs from a regime of weak legal protection that tend to use social and personal networks to protect their property rights may find it difficult to use sophisticated contractual approaches to protect their interests in a strong legal environment.

As for advising, VCs that excel in market-based strategies in a well established institutional environment might find it difficult to provide strategic advice to an investee that relies primarily on network strategies to survive in an underdeveloped institutional environment, and vice versa (Bruton and Ahlstrom, 2003). In addition, VCs that add value to investees through replacement of founders with professional CEOs may not effectively use similar approaches to provide value-added services in countries with weak legal enforceability where the replacement decision cannot be enforced (Ahlstrom and Bruton, 2006).

Such incongruence in practices implies that VCs would encounter significant challenges and incur additional costs conducting primary activities in an institutionally distant environment. Moreover, institutional distance increases the risk of legitimacy and reputation loss and lowers VCs' ability to legitimize practices that they attempt to introduce to their investee ventures (Kostova and Zaheer, 1999). Institutional distance increases the likelihood that some well accepted practices in one institutional environment are not perceived as legitimate in another environment (Kostova and Zaheer, 1999). In the venture capital context that is already characterized by information asymmetry and uncertainty (Amit et al., 1998; Li, 2008; Ruhnka and Young, 1991), lack of legitimacy and reputation will constrain the number of investment deals VCs can access and thus reduce VCs' effectiveness in deal selection. In addition, investees will have reduced motivation to adopt the practices recommended by VCs that are perceived as less legitimate, which will further hamper the effectiveness of coaching and advising (Kostova and Zaheer, 1999).

In brief, since institutional distance causes incongruence in practices and reduces the legitimacy of VCs' knowhow and practices in the host environment, it will decrease VCs' effectiveness in conducting their primary activities and hamper their international investment performance. We thus propose the following hypothesis.

Hypothesis 2. The larger the institutional distance between VCs and investee ventures, the lower the likelihood of exit success in international venture capital investments.

3.5. Cross-border investment experience, national distances and venture capital investment performance

Accumulation of experience is one of the most important sources of organizational learning (Luo and Peng, 1999). Organizational learning is the process through which organizations "encode influences from history into routines that guide behaviors" (Levitt and March, 1988: 320). Learning from cross-border investment experience, VCs can accumulate knowledge,

develop organizational routines, and build relational capital to mitigate the negative impacts of institutional and cultural distances on investment performance. The nature, intensity, and outcomes of experiential learning depend, however, on the types of experience and the way in which the organization encodes experience. We consider two types of cross-border experience: broad international experience and host country-specific experience. Broad international experience reflects the breadth and scope of experience in terms of the diversity of countries in which a VC has invested, while host country experience represents the depth or intensity of investment in a specific destination.

Broad international experience can mitigate the negative consequences of institutional and cultural distances in two ways. At the very least, such experience enhances VCs' awareness, tolerance, and appreciation of cross-country differences and thus increases VCs' motivation to develop knowledge and capabilities to deal with country differences (Delios and Beamish, 2001; Delios and Henisz, 2003; Johanson and Vahlne, 1977). Although exposure to different cultural environments does not necessarily decrease the gap in values or cognitive structures, VCs can learn through such experience to appreciate cultural differences and increase their tolerance and sensitivity to unexpected behaviors and practices.

Broad international experience can similarly increase VCs' awareness of differences in the institutional rules between home and host countries, and enable VCs to accumulate knowledge related to diverse institutional practices (Johanson and Vahlne, 1977). These differences include those in the interpretation and enforcement of rules as well as those in business practices that are not codified but are taken for granted. Further, since VCs are likely subject to legal or social sanctions for behaviors incompatible with host institutional environments, experienced VCs can quickly discover the importance of compliance with local institutions and develop standard operating procedures to adapt to laws, rules, and processes as well as negotiate with authorities to legitimize actions and practices that do not yet conform to existing rules.

More important, broad international experience increases VCs' adaptive capability to bridge the gaps created by national distances (Meuleman and Wright, 2011). Having investment and operational experience in a variety of countries exposes VCs to new events and ideas (Luo and Peng, 1999), which facilitates the development of mental models attuned to cope with broader ranges of challenges and environmental changes (Lavie and Miller, 2008). While knowledge learned in other countries might not apply directly to the host country, VCs with experience of investing in a wide variety of countries develop decision heuristics such as analog reasoning (Dimov and de Holan, 2010) that allow them to screen, reinterpret, and adjust such knowledge to a new context (Zahra and George, 2002). Broad experience also makes it more likely to find a close match for a problem in a new geographical domain, or for a new problem in a host country where changing conditions make obsolete host country specific knowledge accumulated in the past (De Clercq et al., 2005; Dimov and de Holan, 2010). Further, broad international experience helps develop translation capability, the ability to translate concrete practices developed at home or in other countries into context free abstractions and then unpack the abstractions into suitably modified practices in the host country (Dokko and Gaba, 2012).

As broad international experience enhances VCs' awareness of hazards related to unfamiliarity with local institutional and cultural environments and increases VC's capability of adapting practices to host country environments, it will improve VCs' abilities to deal with institutional and cultural differences (de Clercq and Dimov, 2008), when conducting primary venture capital investment activities (Lavie and Miller, 2008). Therefore, we formally propose that:

Hypothesis 3a. VCs' broad international experience will attenuate the negative effect of cultural distance between VCs and investee ventures on the likelihood of exit success in international venture capital investments.

Hypothesis 3b. VCs' broad international experience will attenuate the negative effect of institutional distance between VCs and investee ventures on the likelihood of exit success in international venture capital investments.

Host country-specific experience, as compared with broad international experience, allows for deep understanding of host country institutional and cultural environments and thus acquisition of knowledge about the differences in the environments. Such understanding and knowledge enhance opportunity assessment and transfer of knowledge and practice routines to the specific host country. Deep knowledge of the host country environment helps VCs identify knowledge overlaps, which, in turn, improves the disseminative capacity of VCs (Minbaeva and Michailova, 2004) and the absorptive capacity of investees (Cohen and Levinthal, 1990). These improvements result in better communications between VCs and investees and lower information and transaction costs. In addition, deep understanding of the host country environment facilitates customization of the practices VCs bring to the host country (Dokko and Gaba, 2012).

Host country experience can also help VCs build relationships and develop skills to overcome relational impediments and obtain legitimacy (Kostova and Zaheer, 1999). In particular, a long established presence may contribute directly to increased legitimacy of VCs in the host country (Child, 1994), provide VCs with opportunities to develop practices that better fit the host institutional environment (Luo and Peng, 1999), and develop skills to manage its legitimacy through negotiation with the host institutional environment (Kostova and Zaheer, 1999). We thus propose that:

Hypothesis 4a. VCs' host country experience will attenuate the negative effect of cultural distance between VCs and investee ventures on the likelihood of exit success in international venture capital investments.

Hypothesis 4b. VCs' host country experience will attenuate the negative effect of institutional distance between VCs and investee ventures on the likelihood of exit success in international venture capital investments.

We further posit that closing national distance gaps is easier in those domains that are explicit and involve coercive “feedback” when actions taken do not comply with institutional restrictions, and in those domains where adaptation can be more readily made. These domains are constituted largely of formal institutions that are codified in laws and regulations as well as in salient informal practices. Institutions are typically enforced through coercive methods, and thus behaviors incompatible with them are likely to be punished through specialized enforcement organizations (e.g., court and the police) (Scott, 2001). Given the action environment and task domains of VCs, they must adapt to the host country’s formal institutional rules and salient business practices to obtain and maintain legitimacy. Legal, regulatory and other sanctions imposed in the case of failed formal institutional compliance raise VCs’ awareness of institutional differences across nations, and increase the need to adapt to such differences. Similarly, engagement in business practices and behaviors that do not fit the socially legitimate ways of doing business in the host country, even if not formalized, is likely to result in visible social sanctions (e.g. shunning, exclusion, and other forms of sanctions), which in turn propel VCs to learn through experience.

In contrast, the elements of cultural distance that mostly affect VCs’ interactions with investees and other stakeholders in the host country environment are largely tacit, socially constructed over time, and reflected more in deeply held values and programmed cognitive patterns (Hofstede, 1989). The consequences of cultural distance are therefore more subtle and often subconscious, which makes them more difficult to identify and tackle (Hofstede, 1989; North, 1990). In fact, each party to the interaction may not be entirely aware of the conventions and codes it follows in prioritizing, perceiving, interpreting, and acting, or it may assume that these conventions and codes are part of a natural factual order that is taken for granted (Hofstede, 2001; Schein, 1985; Scott, 2001). Furthermore, since values are often unconsciously held, the effects of cultural distance, such as the liabilities of outsidership, are likely to persist and cannot be easily eliminated by conscious effort (Hofstede, 1989). Barkema and Vermeulen (1997), for instance, found that the effect of cultural distance on the survival of international joint ventures was consistently negative over the 18-year period in their sample.

Thus, the consequences of national institutional distance are likely more visible and identifiable than those of cultural distance in international interactions. Such explicit and salient nature of institutional differences permits organizations to obtain more information and to decide more effectively how to cope with any deleterious effects resulting from institutional distance. We therefore expect that the closure of the national distance gap through both broad international and host country experiences will be more salient in adaptation to institutional distance than in adaptation to cultural distance.

It is worth noting that the value of knowledge gained from additional international experience may start declining at some point when knowledge gaps are closed and thus opportunities for learning are exhausted. Since cultural knowledge is more tacit and slower to acquire than institutional knowledge, the marginal value of learning from experience about culture may at some point exceed the marginal value of learning about institutions where learning opportunities are likely exhausted earlier. In the venture capital context, however, VCs’ interaction with foreign investees and their environments tends to be episodic as compared with other forms of foreign direct investment such as wholly owned subsidiaries. Further, the dynamic nature of institutional environments often provides new learning opportunities. For these reasons, we suggest that international and host country experiences will have more value in closing institutional rather than cultural knowledge gaps.

Hypothesis 5. Broad international and host country experiences will have more salient mitigating effects on the negative relationship between institutional distance and exit success (H3b and H4b) than on the negative relationship between cultural distance and exit success (H3a and H4a) in international venture capital investments.

4. Methods

4.1. Data and sample

This study focuses on how national distances influence venture capital investment performance. We collected venture capital investment data from the VentureXpert database, which has been used extensively in existing research (Dai et al., 2012; Hochberg et al., 2007; Li, 2008; Shane and Stuart, 2002; Sorenson and Stuart, 2001).

We constructed our sample as follows. First, we collected standard venture capital investments in which the investee venture has received funding from a foreign VC, defined as one whose country of origin is different from the investee’s. We focused on investments in the first financing round, because VCs make important strategic decisions such as deal evaluation and selection in the first round, and because their interaction with investee ventures in this initial round lays the foundation for subsequent exchange and is critical for the investment’s ultimate success (De Clercq et al., 2006; Fitza et al., 2009; Sahlman, 1990). We included investments that received initial funding between 1996 and 2008. Globalization of venture capital has gathered pace only since the mid- to late-1990s (Iriyama et al., 2010; Kenney et al., 2008). We tracked the outcome of each investment until the end of 2011 and had at least three years to observe an exit for an investment initiated as late as in 2008 (Gompers and Lerner, 1998; Hochberg et al., 2007; Nahata, 2008). Our sample consists of 10,593 international venture capital investments, for which relevant venture, VC and country information is available. In the sample, 7816 VCs from 69 nations invested in 10,593 ventures from 64 nations, and 5993 VCs made cross-border investments; 6520 ventures received investments from foreign VCs only, and about 38% from a syndicate of foreign and local VCs.

4.2. Dependent variable

It is a dummy that takes the value one if the investee venture ultimately went public or was acquired. Data constraints prevented us from computing project-specific rates of return. Given that VCs generate returns primarily from successful exits

through either venture IPO or sale (NVCA, 2008; Triantis, 2001), we view an exit in these two forms as a final indicator of investment success. A large number of recent studies have adopted the same measure of venture capital investment success (e.g., Bayar and Chemmanur, forthcoming; Bottazzi et al., 2008; Cumming and Dai, 2010; Dai et al., 2012; Nahata, 2008; Zarutskie, 2010). Although IPO is more profitable (Brander et al., 2002; Gompers and Lerner, 2000; Martens et al., 2007; Sorenson and Stuart, 2001), venture sale on average remains an attractive exit option as well (Guler, 2007; Hochberg et al., 2007). We verified the acquisition and IPO data through the SDC Global New Issues and M&A databases provided by Thomson Financial.

4.3. Explanatory variables

4.3.1. Institutional distance

We generated the measure in two steps. First, we collected the rank measures of governance institutions from the 2011 World Governance Index (WGI) constructed by Kaufmann et al. (2007). The WGI is built on hundreds of specific and individual measures, taken from 33 data sources provided by 30 different organizations. It covers almost all the countries and territories in our sample, thus minimizing data loss. Though measurement error is pervasive in indicators of institutions and governance, a key advantage of the WGI is that the indicators are more informative about unobserved governance and have “substantially smaller [margins of error] than any of the individual data sources” (Kaufmann et al., 2007: 1). The WGI includes six dimensions: government effectiveness (e.g., quality of bureaucracy), quality of institutional policies (concerning competition, banking/finance, taxation, and trade), rule of law (including property rights protection), political stability, voice and accountability (e.g., public access to information and freedom of expression), and control of corruption (Kaufmann et al., 2007). Given that the individual dimensions are highly correlated (with pairwise correlations ranging between 0.69 and 0.93), we developed a composite index by averaging the six dimensions. Second, we computed the Manhattan (absolute) distance between each VC–venture pair based on the composite index. In case of multiple co-investors, we used the equally weighted average distance in recognizing the fact that collective decisions are made by all co-investors in the first (strategic) round.⁴ This measure is time varying.

4.3.2. Cultural distance

We generated the Kogut and Singh (1988) cultural distance measure between a VC and a venture based on the values of Hofstede's (2001) four cultural dimensions: uncertainty avoidance, individualism/collectivism, power distance and masculinity/femininity. Despite the acknowledged limitations (Shenkar, 2001), the Hofstede cultural dimensions have been employed extensively in studies on international business (for a review, see Tihanyi et al., 2005), innovation, and entrepreneurship (e.g., Shane, 1992, 1995; Steensma et al., 2000). Current research has yet to offer a better measure of national cultural distances (Lavie and Miller, 2008). We used the equally weighted average distance in case of multiple co-investors. This measure is time invariant.

4.3.3. Broad international experience

It is measured as the amount of foreign investments a VC has made in the past five years normalized by the total amount of investments the VC has made in the past five years. This measure is time varying and VC specific. In case of multiple co-investors, we used an equally weighted average measure. It ranges from zero (least international) to one (most international).

4.3.4. Host country-specific experience

We measured host country experience as the total number of ventures a VC has funded in the host country in the past five years normalized by the total number of ventures the VC has funded in the past five years. In case of multiple co-investors, we used an equally weighted average measure. This measure ranges from zero to one. It is time varying and VC specific.

4.4. Control variables

The performance of an investment project is likely influenced by other factors. We first included several variables to control for VC characteristics. Investment performance might be driven by unobservable VC-specific characteristics, which should be captured by *VC past success*. It is the number of ventures funded by a VC that have gone public or been sold in the past five years. We took the log of the measure. Since VCs tend to be industry specialized (Barry, 1994; Cumming, 2006; Gupta and Sapienza, 1992; Norton and Tenenbaum, 1993; Sorenson and Stuart, 2001), we also controlled for *VC industry specialization*. It is the amount of investment (in thousand US\$) in the focal venture's industry by a VC in the past five years, normalized by the total amount of investment by the VC in the past five years. *VC syndication* can help VCs select deals and enhance monitoring and advising capabilities, thus contributing positively to investment performance (Lerner, 1994). We included a dummy variable that takes the value one if a VC is joined by co-investors in a deal. Finally, *VCs with local offices* may have better access to information about the

⁴ It is worth noting that the investment amount of each VC in each venture is either missing or incomplete for about a quarter of our sample observations, which precludes us from generating a value-weighted distance measure. We also considered an alternative measure, i.e., the minimum distance between syndicate partners and the investee. This approach, however, suggests delegating decisions to the host country members who are familiar with the local environment and local practices. This approach is more appropriate in the operational stage where host country members can be used to bridge differences.

local context and help reduce the negative consequences of national distances. It is a dummy variable that takes the value one if a VC or its co-investor has a branch office in the host country. We collected information about branch offices primarily from *Galante's Venture Capital & Private Equity Directory* and *Pratt's Guide*, and from *VentureXpert* in case of missing values.

Second, we controlled for venture-related characteristics. *Early-stage ventures* are less likely to achieve significant sales or returns (NVCA, 2008) and might have a higher likelihood of failure. We thus included a dummy variable that takes the value one if, at the time of the initial financing round, the venture is in the start-up/seed and early stages of development, and zero if in the expansion and later stages. We collected the stage of development information from *VentureXpert*. In addition, we accounted for venture industry-specific fixed effects by including *industry dummies* in our estimations. We reclassified the VEIC codes (Venture Economics Industry Classification) into 49 industries developed by Fama and French (1993) (see also Li and Mahoney, 2011).

Third, since geographic distance increases information gathering and monitoring costs (Sorenson and Stuart, 2001) and decreases the effectiveness of knowledge transfer between organizations (Gilbert et al., 2008), greater *geographic distance* may negatively affect venture capital investment performance (Cumming and Dai, 2010; Dai et al., 2012). This measure is the distance between the VC's location and the investee venture's location, computed in nautical miles (Sorenson and Stuart, 2001). We collected the latitude and longitude information for the city of each VC and venture from the GeoDataSource World Cities Database (Gold Edition), and if missing, for the state/provincial capitals, and in a few cases, for the national capitals. We took the log of this measure.

Fourth, we included two variables to control for venture nation time-varying effects on investment performance. Higher venture capital commitments in a nation (*Venture nation VC commitment*) may lead to more funding for some investee ventures and increase their chances of survival, but more capital commitments to venture capital also allow VCs to pursue more promising projects and, in this process, to abandon some existing projects. This time-varying measure is the total amount of funds raised by VCs in a nation (in million US\$), scaled by national GDP (in thousands). We added one to this measure and took the log of it. Nations with higher GDP (*Venture nation GDP*) might have more entrepreneurial potential. On the other hand, as VCs rush to such bigger markets, competition might dampen the probabilities of success. In addition, we included *venture-nation dummies* and *year dummies* (the last date of investment prior to exit, i.e., 1996, 1997, ..., 2011) to control for venture country fixed effects and unobservable temporal effects.

4.5. Estimation models

We estimated both logit and Heckman models. Since our dependent variable, the likelihood of exit success, is binary in nature, we applied the logit model (Greene, 2002):

$$P(y = 1|x) = F(z) = \frac{\exp(z)}{1 + \exp(z)}$$

where $P(y = 1|x)$ is the probability of a successful exit conditional on the set of explanatory variables x , and $F(z)$ is the cumulative distribution function for a standard logistic random variable z , which, in turn, is a linear function of the explanatory variables: $z = \mathbf{x}'\boldsymbol{\beta}$. The model is estimated with the maximum likelihood approach. Following Ai and Norton (2003), we computed the interaction effect of two variables, x_1 and x_2 , as

$$\text{InterEffect} = \frac{\partial^2 F}{\partial x_1 \partial x_2} = \beta_{12}F(1-F) + (\beta_1 + \beta_{12}x_2)(\beta_2 + \beta_{12}x_1)F(1-F)(1-2F).$$

Selection bias can potentially affect the logit estimation, since we included an investment only if it involved at least a foreign VC. Essentially, we observed the performance of a cross-border investment only when a foreign investor participated in the deal, but we did not observe the “counterfactual outcome” when no foreign investor participated. In addition, entrepreneurial ventures with different performance may differ in their propensity to invite a foreign investor (Dai et al., 2012). In the absence of a natural experiment, we specified a two-stage Heckman model (Heckman, 1979). In the main regression equation, we used a probit model to estimate the likelihood of an international venture capital investment having a successful exit ($y_j = 1$):

$$y_j = \mathbf{x}_j + u_{1j}.$$

Since we observe the exit success of an international venture capital investment conditional on the deal having at least a foreign VC, we used a probit model in the selection equation to estimate the likelihood of an investment having a foreign investor:

$$z_j\gamma + u_{2j} > 0$$

where $u_1 \sim N(0, \sigma)$, $u_2 \sim N(0, 1)$ and $\text{corr}(u_1, u_2) = \rho$. $\mathbf{Z}$ is the vector of variables used to explain the likelihood of an international venture capital investment.

We obtained a “control” group and merged it with our sample in the Heckman estimation. From the *VentureXpert* database, we collected all domestic venture capital investments during 1996–2008. We then excluded those deals involving domestic

investors that have simultaneously co-invested with foreign VCs in the host country (i.e., domestic investors that appeared in our sample in the same year) to avoid any confounding effects. The control group has 16,419 observations.

We chose the sectoral propensity to have a foreign investor in the past five years as the “identification” variable in the selection equation. On the one hand, the sectoral propensity should be positively related to the propensity for a specific deal (the relevance condition); on the other hand, the sectoral variable should be largely exogenous to the performance of a specific deal in a nation, after we control for industry fixed effects, country fixed effects, and other factors (the exclusion criterion). In addition, we included the variable *early-stage ventures*—an indicator of the venture's progress in terms of its commercial potential, because a later stage of development by definition implies establishment of a stable consumer base or even revenue generation and profitability (Dushnitsky, 2004; NVCA, 2008). We also accounted for the effects of syndication and venture nation investment opportunities.

5. Results

Table 1 presents the descriptive and correlation statistics for the variables. It shows that exit success is negatively correlated with both institutional and cultural distances. The variance inflation factors (VIFs) are well below the recommended threshold of 10 for multicollinearity (Chatterjee et al., 2000).

5.1. Logit regression results

Table 2 reports the logit estimation results. Model 1 is the baseline model with constant and controls only, Models 2–9 examine the effects of the explanatory variables, and Models 10–11 are for robustness tests. The AIC and McFadden's R-squares in Table 2 suggest that Model 5 is the best fitted, and we used it to gage the economic effects of the explanatory variables.

Hypothesis 1 suggests that the likelihood of a successful exit will decrease as cultural distance between VCs and ventures increases. As expected, the coefficient estimate of cultural distance is negative and statistically significant in Model 4. We computed the “elasticity” of exit success with regard to cultural distance in Model 5. Holding the value of all other variables at the mean level, for one percent increase in the level of cultural distance, the percentage change in the probability of exit success is a decrease of 0.074.

Hypothesis 2 predicts that institutional distance between VCs and investees adversely affects venture capital exit success. The coefficient of institutional distance is negative and significant in Model 4, providing support for H2. For one percent increase in the level of institutional distance, the percentage change in the probability of exit success is a decrease of 0.067. Overall, inclusion of cultural and institutional distances in Models 2–4 results in a significant increase in their explanatory power over Model 1, as reflected by the significant incremental improvements in log-likelihoods ($p < 0.001$).

Hypothesis 3a posits that VCs' broad international experience will attenuate the negative consequences of cultural and institutional distances between VCs and ventures. In Model 5, while the interaction term of cultural distance with VC broad international experience has a statistically insignificant coefficient estimate, the interaction term of institutional distance with broad international experience has a statistically significant and positive coefficient. Incorporating the moderating effect of broad international experience moderately improves the explanatory power of Model 5 relative to Model 4 ($p < 0.09$). Thus, H3 is partially supported.

Hypothesis 4a suggests that VCs' host country-specific experience can mitigate the negative effects of national distances. In Model 6, the coefficient estimates of the interaction terms of both institutional and cultural distances with VC host country experience are statistically insignificant. Adding the interaction terms does not improve the explanatory power of the model relative to Model 4 ($p < 0.94$). H4a and H4b are not supported.

Fig. 1a shows the magnitude and sign of the moderating effect of VC broad international experience with 90% confidence intervals. The negative impact of institutional distance on VC exit success decreases as broad international experience increases. Since the interaction effect of institutional distance varies not only with the value of broad international experience but also with the value of all other covariates in the logit estimation (Ai and Norton, 2003; Huang and Shields, 2000), we further analyze how the predicted probability of exit success changes with the level of broad international experience at different levels of covariate contribution (CC) (i.e., the composite influence of all the other covariates) (Hoetker, 2007; Mitchell and Chen, 2005). Fig. 1b shows a similar pattern as Fig. 1a, although the moderating effect of international experience decreases as the covariate contribution increases.

Hypothesis 5 posits that VCs' broad international and host country experiences will have more salient mitigating effects on the negative impact of institutional distance than on the negative impact of cultural distance. Because the logit model is nonlinear and the units of measurement differ for institutional and cultural distances, it is inappropriate to compare simply the coefficients of the interaction terms, but we can compare their sign and significance (Wooldridge, 2006). For this purpose, we included the interaction terms of cultural and institutional distances in Models 8 and 9, respectively. While broad international and host country experiences have no significant mitigating effects on the negative consequences of cultural distance in Model 8, broad international experience has a significant moderating effect on the negative consequences of institutional distance in Model 9. We also compared the explanatory power of Models 8 and 9 with that of Model 4 which has no interaction terms. We found that Model 9 significantly increases the incremental log-likelihoods ($p < 0.09$) but Model 8 does not. Thus, H5 is partially supported.

Table 1Descriptive and correlation statistics for the logit estimation sample. ^{a,b}

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
(1) Exit success	1.00															
(2) VC past success (logged)	0.06	1.00														
(3) VC industry specialization	0.00	−0.24	1.00													
(4) VC syndication	0.05	0.01	0.28	1.00												
(5) VC with local office	0.04	0.06	0.05	0.21	1.00											
(6) Early-stage ventures	−0.06	−0.05	0.22	0.19	0.10	1.00										
(7) Geographic distance (logged)	−0.02	0.06	0.03	−0.09	−0.08	−0.03	1.00									
(8) Venture nation VC commitment (logged)	0.01	0.01	−0.01	−0.02	0.02	−0.02	0.05	1.00								
(9) Venture nation GDP (logged)	0.01	0.10	0.07	0.15	0.17	0.13	0.04	−0.06	1.00							
(10) VC broad international experience	0.03	−0.09	0.32	−0.08	−0.03	−0.04	0.02	0.02	−0.08	1.00						
(11) VC host country experience	0.00	−0.10	0.13	0.47	0.30	0.17	−0.15	−0.04	0.31	−0.04	1.00					
(12) Cultural distance	−0.05	0.08	−0.15	−0.30	−0.15	−0.14	0.32	0.08	−0.22	0.10	−0.32	1.00				
(13) Institutional distance	−0.04	0.08	−0.16	−0.27	−0.04	−0.15	0.34	0.00	−0.18	0.13	−0.28	0.51	1.00			
(14) Sectoral propensity to have foreign investors ^c	−0.06	−0.01	−0.05	0.03	−0.18	−0.04	0.10	0.03	−0.15	0.16	−0.11	0.15	0.17	1.00		
(15) Language distance ^d	−0.05	0.07	−0.22	−0.43	−0.17	−0.20	0.14	0.06	−0.18	0.16	−0.43	0.67	0.41	−	1.00	
(16) Legal distance	−0.05	0.09	−0.16	−0.25	−0.03	−0.15	0.30	0.01	−0.17	0.10	−0.26	0.50	0.93	−	0.41	1.00
Mean	0.31	0.75	0.48	0.57	0.22	0.42	8.15	0.01	27.74	0.50	0.31	1.44	13.16	0.27	−1.79	0.48
S.D.	0.46	0.49	0.33	0.50	0.41	0.49	1.07	0.11	1.54	0.33	0.38	1.35	14.91	0.09	1.65	0.61
VIF	−	1.12	1.41	1.48	1.14	1.10	1.23	1.02	1.21	1.21	1.55	1.54	1.53	−	−	−

^a N = 10,593 but see notes ^c and ^d.^b Correlations with absolute values larger than 0.03 are significant at $p < 0.01$.^c Correlations for the sample of 27,012 used in the Heckman estimation.^d Correlations for language distance (N = 10,578).

Table 2

Logit model results for the likelihood of exit success.

	1	2	3	4	5	6	7	8	9	10	11
VC past success (logged)	0.126** (0.0466)	0.142** (0.0467)	0.135** (0.0466)	0.144** (0.0467)	0.154** (0.0470)	0.145** (0.0468)	0.154** (0.0471)	0.145** (0.0468)	0.154** (0.0470)	0.145** (0.0471)	0.156*** (0.0471)
VC industry specialization	−0.118 (0.0823)	−0.155 + (0.0827)	−0.158 + (0.0827)	−0.174* (0.0829)	−0.168* (0.0829)	−0.173* (0.0831)	−0.167* (0.0831)	−0.173* (0.0830)	−0.168* (0.0831)	−0.199* (0.0838)	−0.168* (0.0831)
VC syndication	0.247*** (0.0529)	0.209*** (0.0534)	0.216*** (0.0533)	0.196*** (0.0536)	0.198*** (0.0537)	0.197*** (0.0537)	0.198*** (0.0537)	0.195*** (0.0536)	0.198*** (0.0537)	0.181*** (0.0545)	0.197*** (0.0537)
VC with local office	0.176** (0.0548)	0.168** (0.0548)	0.203*** (0.0551)	0.190*** (0.0552)	0.183*** (0.0553)	0.190*** (0.0555)	0.184*** (0.0556)	0.189*** (0.0553)	0.182*** (0.0554)	0.180** (0.0556)	0.193*** (0.0557)
Early-stage ventures	−0.393*** (0.0480)	−0.400*** (0.0481)	−0.400*** (0.0481)	−0.403*** (0.0481)	−0.401*** (0.0481)	−0.402*** (0.0481)	−0.401*** (0.0481)	−0.402*** (0.0481)	−0.401*** (0.0481)	−0.408*** (0.0483)	−0.403*** (0.0482)
Geographic distance (logged)	−0.0207 (0.0208)	0.0161 (0.0220)	0.0177 (0.0222)	0.0341 (0.0227)	0.0374 (0.0227)	0.0333 (0.0228)	0.0365 (0.0229)	0.0341 (0.0228)	0.0373 (0.0228)	0.0236 (0.0225)	0.0383 + (0.0227)
Venture nation VC commitment (logged)	0.177 (0.195)	0.230 (0.196)	0.145 (0.196)	0.195 (0.197)	0.193 (0.197)	0.197 (0.197)	0.195 (0.197)	0.203 (0.197)	0.193 (0.197)	0.174 (0.197)	0.201 (0.197)
Venture nation GDP (logged)	−0.000138 (0.0156)	−0.0127 (0.0158)	−0.0115 (0.0158)	−0.0179 (0.0159)	−0.0177 (0.0160)	−0.0180 (0.0159)	−0.0179 (0.0160)	−0.0189 (0.0160)	−0.0177 (0.0159)	−0.0130 (0.0159)	−0.0186 (0.0159)
VC broad international experience	0.281*** (0.0719)	0.322*** (0.0724)	0.336*** (0.0729)	0.352*** (0.0730)	0.213 + (0.110)	0.348*** (0.0741)	0.211 + (0.110)	0.276** (0.105)	0.211* (0.0974)	0.328* (0.132)	0.230* (0.108)
VC host country experience	−0.162* (0.0710)	−0.209** (0.0717)	−0.204** (0.0716)	−0.226** (0.0719)	−0.236** (0.0721)	−0.248** (0.0948)	−0.255** (0.0948)	−0.249** (0.0922)	−0.239** (0.0868)	−0.515** (0.174)	−0.257** (0.0936)
Cultural distance		−0.103*** (0.0192)		−0.0781*** (0.0206)	−0.0720 + (0.0378)	−0.0814*** (0.0234)	−0.0754 + (0.0387)	−0.108** (0.0351)	−0.0732*** (0.0207)	−0.0646* (0.0290)	−0.0678 + (0.0382)
Institutional distance			−0.00883*** (0.00176)	−0.00622*** (0.00189)	−0.0122*** (0.00367)	−0.00622** (0.00214)	−0.0121** (0.00372)	−0.00609** (0.00189)	−0.0122*** (0.00340)	−0.0129*** (0.00345)	−0.339*** (0.0883)
Cultural distance × VC broad international experience					−0.00215 (0.0546)		−0.00414 (0.0550)	0.0472 (0.0485)		0.0394 (0.0430)	−0.00256 (0.0542)
Institutional distance × VC broad international experience					0.00992* (0.00506)		0.0101* (0.00509)		0.00981* (0.00449)	0.00835 + (0.00476)	0.238 + (0.123)
Cultural distance × VC host country experience						0.0204 (0.0670)	0.0281 (0.0673)	0.0189 (0.0572)		−0.0744 (0.0498)	0.0248 (0.0662)
Institutional distance × VC host country experience						0.000169 (0.00682)	−0.00116 (0.00688)		0.000357 (0.00583)	0.00620 (0.00650)	−0.0311 (0.171)
Constant	0.181 (0.592)	0.380 (0.595)	0.357 (0.595)	0.462 (0.596)	0.504 (0.600)	0.481 (0.599)	0.525 (0.602)	0.534 (0.601)	0.507 (0.598)	0.289 (0.599)	0.521 (0.602)
Observations	10,593	10,593	10,593	10,593	10,593	10,593	10,593	10,593	10,593	10,578	10,593
Chi square	599.1	628.3	624.8	639.3	644.2	639.5	644.4	640.4	644.2	645.9	653.0
AIC	12,700	12,673	12,676	12,664	12,663	12,668	12,667	12,667	12,663	12,658	12,658
McFadden's R ²	0.0455	0.0477	0.0475	0.0486	0.0489	0.0486	0.0489	0.0486	0.0489	0.0491	0.0496
Log likelihood	−6283	−6268	−6270	−6263	−6261	−6263	−6260	−6262	−6261	−6256	−6256
Compared to model		1	1	1	4	4	5	4	4		
Change in likelihood (d.f.)		29.24 (1)	25.65 (1)	40.22 (2)	4.849 (2)	0.138 (2)	0.177 (2)	1.084 (2)	4.851 (2)		
Significance of improvement (p value of LR test)		6.40e−08	4.10e−07	1.84e−09	0.0885	0.933	0.915	0.582	0.0884		

Notes: This table reports the results of logit models for the likelihood of exit success. Variables are defined in the text. Venture industry dummies, nation dummies, and year dummies are included but not reported. Models 1–9 are estimated with the cultural and institutional distance measures, Model 10 with language distance, and Model 11 with legal distance. Standard errors appear in parentheses. ***p < 0.001, **p < 0.01, *p < 0.05, + p < 0.10 (two-sided tests). All Chi square tests are significant at p < 0.001.

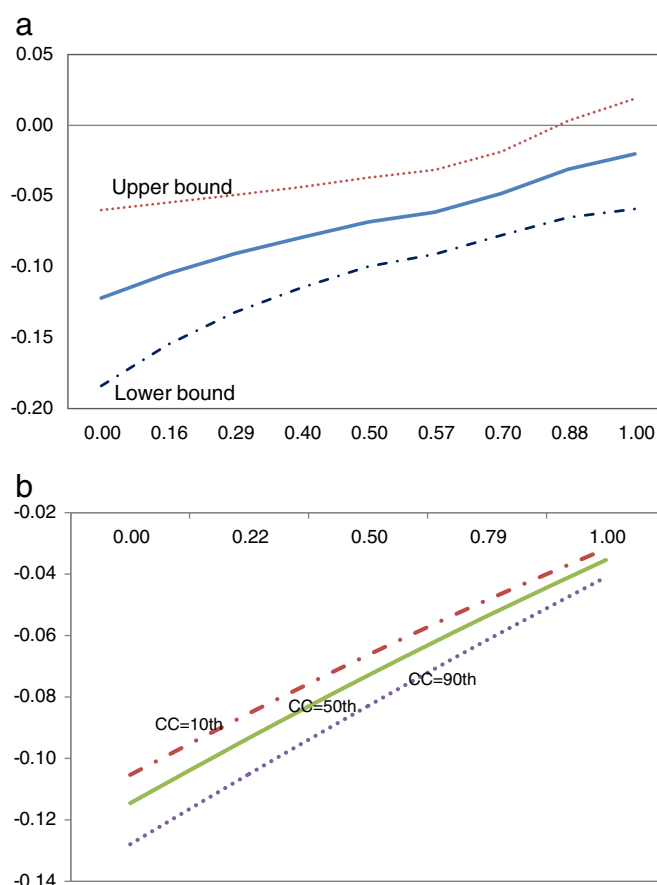


Fig. 1. Interaction effect of institutional distance and broad international experience. Notes: The figures are based on the logit estimates in Model 5 of Table 2. The y-axis represents the percentage change in the probability of exit success due to a one percent change in institutional distance; the x-axis represents the values of broad international experience. Upper bound and lower bound are the 90% confidence levels for the percentage change in the probability of exit success. CC = 10th, 50th, 90th indicate the contributions of the covariates other than institutional distance and broad international experience, with the covariates' values at the 10th, 50th, 90th percentiles, respectively.

Several empirical findings about the control variables in Table 2 are noteworthy. Both VCs' past success and VC syndication help improve the success rate of the investment project, as does the presence of local branch offices. On the other hand, VCs that are specialized in the venture industry have a lower likelihood of exit success in the international venture capital context. As expected, ventures that are in their earlier stages when first funded are less likely to experience a successful exit.

5.2. Heckman regression results

Table 3 presents the Heckman estimation results. Model 4 includes both institutional and cultural distances; Models 5 and 6 the interaction terms of broad international experience and host country experience, respectively; Model 7 all the interaction terms. The results for our explanatory variables are consistent with those of the logit estimations. For the selection equation, the likelihood of having an international venture capital investment is positively associated with the sectoral propensity, VC syndication, and the level of venture capital commitment in the host nation, but negatively with venture early stage of development and venture nation GDP.

5.3. Additional analyses

We conducted a number of additional analyses to examine the robustness of the results. Given the caveats about Hofstede's cultural dimensions, we employed an alternative measure that is based on the difference between the major languages used in any two nations (Dow and Karunaratna, 2006). Details about the measure can be found at <https://sites.google.com/site/ddowresearch/home/scales>. Language is often considered the mirror of culture (Thomas, 2011). If both the VC and the investee venture come from the same country, we assigned to the pair the lowest possible value of language distance, which indicates that the same language is used in the home and host countries. We computed the equally weighted average distance in case of

Table 3

Heckman selection model results for the likelihood of exit success.

	1	2	3	4	5	6	7
<i>Second stage: likelihood of exit success</i>							
VC past success (logged)	0.0740** (0.0269)	0.0821** (0.0270)	0.0792** (0.0270)	0.0839** (0.0270)	0.0890** (0.0272)	0.0842** (0.0271)	0.0891** (0.0272)
VC industry specialization	−0.0638 (0.0471)	−0.0862 + (0.0474)	−0.0868 + (0.0476)	−0.0973* (0.0477)	−0.0940* (0.0477)	−0.0967* (0.0478)	−0.0939* (0.0478)
VC syndication	0.342*** (0.0476)	0.320*** (0.0483)	0.319*** (0.0486)	0.309*** (0.0488)	0.309*** (0.0489)	0.309*** (0.0489)	0.309*** (0.0490)
VC with local office	0.104** (0.0318)	0.100** (0.0318)	0.120*** (0.0321)	0.112*** (0.0321)	0.108*** (0.0322)	0.112*** (0.0323)	0.108*** (0.0323)
Early-stage ventures	−0.270*** (0.0281)	−0.274*** (0.0281)	−0.273*** (0.0282)	−0.275*** (0.0282)	−0.274*** (0.0282)	−0.275*** (0.0282)	−0.274*** (0.0282)
Geographic distance (logged)	−0.0160 (0.0120)	0.00556 (0.0127)	0.00578 (0.0128)	0.0158 (0.0131)	0.0175 (0.0131)	0.0156 (0.0132)	0.0173 (0.0132)
Venture nation VC commitment (logged)	0.216 + (0.121)	0.245* (0.121)	0.194 (0.121)	0.223 + (0.122)	0.222 + (0.122)	0.224 + (0.122)	0.222 + (0.122)
Venture nation GDP (logged)	−0.0823*** (0.0184)	−0.0898*** (0.0184)	−0.0864*** (0.0186)	−0.0908*** (0.0185)	−0.0905*** (0.0186)	−0.0907*** (0.0185)	−0.0904*** (0.0186)
VC broad international experience	0.151*** (0.0413)	0.175*** (0.0417)	0.183*** (0.0421)	0.192*** (0.0422)	0.119 + (0.0631)	0.191*** (0.0428)	0.119 + (0.0634)
VC host country experience	−0.0965* (0.0409)	−0.122** (0.0412)	−0.120** (0.0413)	−0.132** (0.0414)	−0.137*** (0.0415)	−0.138* (0.0547)	−0.141* (0.0548)
Cultural distance		−0.0591*** (0.0110)		−0.0456*** (0.0118)	−0.0414 + (0.0213)	−0.0460*** (0.0134)	−0.0421 + (0.0218)
Institutional distance			−0.00499*** (0.00102)	−0.00349** (0.00108)	−0.00666** (0.00205)	−0.00355** (0.00122)	−0.00663** (0.00208)
Cultural distance × VC broad international experience					−0.00280 (0.0308)		−0.00335 (0.0311)
Institutional distance × VC broad international experience					0.00530 + (0.00285)		0.00535 + (0.00287)
Cultural distance × VC host country experience						0.00245 (0.0381)	0.00655 (0.0384)
Institutional distance × VC host country experience						0.000403 (0.00387)	−0.000371 (0.00391)
Constant	2.073*** (0.515)	2.192*** (0.515)	2.119*** (0.519)	2.196*** (0.518)	2.214*** (0.520)	2.197*** (0.518)	2.215*** (0.520)
<i>First stage: likelihood of having a foreign investor</i>							
Sectoral propensity to have foreign investors	2.879*** (0.126)	2.877*** (0.126)	2.877*** (0.126)	2.876*** (0.126)	2.875*** (0.126)	2.876*** (0.126)	2.875*** (0.126)
VC syndication	0.880*** (0.0185)	0.880*** (0.0185)	0.880*** (0.0185)	0.879*** (0.0185)	0.880*** (0.0185)	0.879*** (0.0185)	0.880*** (0.0185)
Early-stage ventures	−0.182*** (0.0186)	−0.182*** (0.0186)	−0.182*** (0.0187)	−0.182*** (0.0187)	−0.182*** (0.0187)	−0.182*** (0.0187)	−0.182*** (0.0187)
Venture nation VC commitment (logged)	0.843*** (0.178)	0.842*** (0.178)	0.845*** (0.179)	0.843*** (0.178)	0.843*** (0.178)	0.843*** (0.179)	0.843*** (0.179)
Venture nation GDP (logged)	−0.371*** (0.00592)	−0.371*** (0.00592)	−0.371*** (0.00592)	−0.371*** (0.00592)	−0.371*** (0.00592)	−0.371*** (0.00592)	−0.371*** (0.00592)
Constant	9.076*** (0.252)	9.079*** (0.252)	9.076*** (0.252)	9.078*** (0.252)	9.079*** (0.252)	9.078*** (0.252)	9.078*** (0.252)
Rho	0.401	0.401	0.388	0.391	0.390	0.391	0.389
Log likelihood	−20,452	−20,437	−20,440	−20,432	−20,430	−20,432	−20,430
Likelihood ratio test of independent equations	20.84	20.63	19.18	19.47	19.25	19.37	19.13
Significance (p value)	5.00e−06	5.58e−06	1.19e−05	1.02e−05	1.15e−05	1.07e−05	1.22e−05

Notes: N = 27,012. This table reports the results of Heckman two-stage probit models for the likelihood of exit success. Variables are defined in the text. Venture industry dummies, nation dummies, and year dummies are included but not reported. Standard errors appear in parentheses. ***p < 0.001, **p < 0.01, *p < 0.05, + p < 0.10 (two-sided tests).

multiple co-investors. Table 1 shows that language distance has a high correlation with cultural distance at 0.67. The results for our hypotheses, as shown in Model 10 of Table 2, are consistent with those in Models 1–9.

Given the importance of law and property rights in venture capital investments (Lerner and Schoar, 2005), we performed a robustness test with one specific institutional dimension, the rule of law. Legal distance is highly correlated with our institutional distance measure at 0.93. Not surprisingly, the new results in Model 11 of Table 2 are consistent with those in Models 1–9.

We also took efforts to address one potential alternative interpretation of our analysis. The results may reflect the impact of the quality of the host-country environment on performance rather than the influence of institutional distance between the host and home countries. For example, if the VCs in our sample had, on average, predominantly better institutional environments than the investees, the negative effects of institutional distance could potentially be interpreted as driven by, for example, low-quality

institutions in the investee country rather than by the *difference* between the host and home institutional environments. It is worth noting that we computed institutional distance as an absolute distance measure. In addition, we conducted several robustness checks. First, we compared the scores of institutional and cultural measures for the VC nations and venture nations for each VC–venture pair in our sample. We found that the institutional scores are lower for the VCs than for the ventures for about 40% of the investments, and higher for 60% of the investments; the cultural scores are lower for the VCs than for the ventures for about 43% of the investments, and higher for 57% of the investments. The data, therefore, do not suggest that the institutional and cultural environments of the VCs are predominantly better than those of the investees, or vice versa. Second, we controlled for time-invariant host-country fixed effects by including venture nation dummies in our estimations, and we controlled for time-varying host-country effects by including venture nation VC commitment and venture nation GDP. These variables should pick up some of the different characteristics of host countries. For these reasons, we believe that our results reflect the effects of institutional distance on performance.

6. Discussion and conclusion

There is a growing consensus that cultural and institutional distances are important in economic transactions (Lavie and Miller, 2008). In the international venture capital context, Dai et al. (2012) addressed the effects of cultural distance on international venture capital investment performance. We extend their contribution by adding an institutional distance perspective to investigate how national distances affect the exit performance of VCs in cross-border investments. We found that both the institutional and cultural distances have a significant negative impact on the performance of international venture capital investments. By including the two distinct but complementary perspectives of national distance in our theoretical and empirical analyses, this study provides a fuller account for the impact of national distance on VC performance.

The high costs of institutional and cultural distances have raised the issue of whether VCs can mitigate national distances by gaining cross-border investment experience. We posited that while broad international experience in diverse countries increases awareness of national differences and allows VCs to develop adaptive capabilities to translate experiences gained from one environment to another, host country experience allows for deeper acquisition of institutional and cultural knowledge and the opportunity to perfect VCs' routines in a particular host country. Our empirical analysis shows that neither type of experience mitigates the negative effects of *cultural* distance on performance in a statistically significant way. It is possible in some international business contexts that “deep learning” through immersion in a culture reduces such a gap. However, given the tacitness of deeply embedded values and cognitive patterns that underlie cultural distance, the episodic nature of VCs' engagement in the host countries may not provide sufficient exposure to support deep learning that is required to bridge the culture gap. Moreover, in many cases the perception of dissimilarity and thus the liability of outsidership cannot be completely eliminated because of attributes that are difficult to change (e.g., differences in physical features).

Our finding that broad international experience in diverse countries mitigates the negative consequences of *institutional* distance suggests that institutional distance features more explicit and salient elements (e.g., legal and social sanctions imposed on those who do not comply) and can thus be more easily bridged. Furthermore, we observed that different from broad international experience, host country experience cannot reduce the negative effects of institutional distance. This finding may lend some support to a more general proposition that the development of knowledge through exploration (diversity of experiences) rather than exploitation (repeated marginal extension of knowledge that allows perfection of routines) might offer a stronger basis for development of competitive advantage.

It is also possible that host country information and knowledge acquired through past experience have a short shelf life, especially in countries undergoing rapid changes. Indeed, Gavetti et al. (2005: 697), echoing Levinthal and March (1993), warn that “deep experience in a source setting might be a double-edged sword.” Managers with deep experience in a host country may become overconfident and profoundly convinced that they know what to do; they apply approaches and practices that succeed in other contexts while ignoring negative feedback from the current environment (Gavetti et al., 2005). In contrast, broad international experience may help managers who have strong host country experience develop better routines to detect changes in a context and respond more sensitively to feedback from changing environments. The impacts of host country deep experience and broad international experience as well as the effects of their interaction on the costs of national distances are not well understood and offer a fertile ground for future research.

Our theoretical development and empirical findings have several implications for practice. First, distances matter to performance, but the main challenges VCs face may not arise from physical or geographic distance but from institutional distance (e.g., achieving social legitimacy by adapting to the relevant institutional environment in the host country) and cultural distance (e.g., displaying sensitivity to differences and reducing their effects by adopting behaviors and practices that are congruent to those of the investee managers and other stakeholders in the host country). Our analysis suggests that cultural and institutional distances have a significant negative effect on venture capital performance, whereas geographic distance does not. Increased effectiveness and lower costs of communication systems, as well as increased convenience and reduced costs of travel, have significantly reduced the importance of physical distance. Institutional and cultural distances, however, often lead to differences in individuals' value systems and behaviors, and their negative impact on venture capital investment performance can persist. These findings have implications for VCs' location choice. VCs need to balance between the benefits associated with growth opportunities in a particular host country and the costs arising from differences in institutional and cultural environments.

Second, our analysis suggests that VCs need to be aware of the differences in acquiring institutional and cultural knowledge. We indicated that the episodic nature of cross-border engagements of VCs may not provide sufficient opportunities for deep

learning that is required to bridge cultural gaps. We also found that international experience in diverse countries (but not host country specific experiences) significantly reduces the deleterious effects of institutional distance. These findings have implications for the staffing decision of international VCs. First, when recruiting professional staff members, VCs might want to prioritize the possession of diverse international experience by candidates. Second, VCs must exercise caution when recruiting professionals to acquire host country knowledge. Specifically, VCs need to ensure that candidates can adapt to changes in local environments so that the knowledge they possess remains current.

6.1. Limitations and future research

We address below some of the limitations of our study and identify several areas for future research that may improve and extend our knowledge of international venture capital. Our use of secondary data and econometric methods masks important relationships and the rich context of venture capital interactions. Future research, field studies and surveys in particular, would enrich and complement our study and shed additional lights on how institutional and cultural distances affect primary venture capital activities in cross-border venture capital investments. The use of exit success to measure VCs' investment performance remains another important limitation of our study. Not all IPOs and sales are equally successful. Given data availability, researchers in the future can gather additional financial and accounting information to capture the performance of venture capital projects.

Some scholars have questioned the face validity of the Hofstede cultural dimensions (e.g., Schwartz, 1990). A number of studies have since been undertaken to generate alternative measures (House et al., 2004; Schwartz, 1990; Triandis, 1989, 1993; Triandis et al., 1986), although they all have their own drawbacks (for a review, see Tihanyi et al., 2005). We obtained robustness results using a language distance measure. Nevertheless, since the notion of culture is complex and multi-dimensional (Triandis et al., 1986), it is worthwhile to look into the implications of these new conceptualizations for venture capital and the effects of specific cultural traits on venture capital activities and performance.

Future research can also extend our study by addressing within country variance in institutional and cultural distances and VCs' response to these distances. First, our national-level measures reflect only the dominant institutions and cultures but do not necessarily capture the regional characteristics (e.g., institutional measures for Canada may not reflect fully the institutional development in Quebec). Second, VCs from the same country may differ in their ability to address the same national distances. The resource-based view has long suggested that firms are heterogeneous in terms of their resource endowments and capabilities (Barney, 1991; Penrose, 1959; Teece et al., 1997). While our study focused on how VCs can use their broad international and host country experiences to reduce the negative consequences of institutional and cultural distances, future research might examine the mitigating effects of other firm-specific factors such as human networks and technological capabilities (Iriyama et al., 2010; Zahra et al., 2000).

Our study points to several broad questions about international venture capital. While VCs are drawn to overseas markets for untapped growth opportunities (Deloitte, 2010), VCs often face significant liability of foreignness due to geographic, institutional, and cultural distances. Some important but underexplored questions ensue: How do VCs deal with the tradeoff between the benefits and costs in cross-border investments? Do VCs on average perform better at home or abroad? Furthermore, foreign and local VCs each have their own advantages and disadvantages (Chemmanur et al., 2011; Cumming and Dai, 2010; Dai et al., 2012). How can foreign VCs attract the best local ventures? How do foreign VCs collaborate with local VCs for mutual benefits? Future research along these lines will add significant insights into our understanding of the development and performance of international venture capital.

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